Guanlin LIU

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EDUCATION

Shandong Normal University **Jinan, China**

*Bachelor of Engineering (*Computer Science and Technology)

- **GPA**: 3.94 (Top 10%), holder of First-class Scholarship and Outstanding Student
- **Notable Courses**: Linea Algebra (100), Data Structure (97), Discrete Mathematics (99), Web Development (97), JAVA Programming (95), HCI Technology (99), Software Engineering (96), Artificial Intelligence (93), Machine Learning (93), Algorithms Analysis and Design (90), Digital Image Processing (92)
- **Graduation thesis**: "Dynamic Graph and Multi-view Drug Information for Safe Medication Recommendation" (mark: 95, supervisor: Prof. Xiaomei Yu)

PUBLICATIONS

- **G. Liu** et al. (2026). ENCMed: A 3D-Encoder Framework with Neighbor Contrastive Learning for Safe Medication Recommendation. [Under Review]
- Z. Liu, H. Wu, X. Fang, **G. Liu** et al. (2026). CLEAR: Prototype-conditioned flow purification for LLM-based rumor detection with Dirichlet evidential learning. *Information Processing & Management*. (IF=8.1) https://doi.org/10.1016/j.ipm.2026.104887
- **G. Liu** et al. (2025). DNMDR: Dynamic Networks and Multi-view Drug Representations for Safe Medication Recommendation. *Knowledge-Based Systems*. (IF=8.0) https://doi.org/10.1016/j.knosys.2025.114327
- X. Li, X. Yu, **G. Liu** et al. (2024). Transformer-based medication recommendation with a multiple graph augmentation strategy. *Expert Systems with Applications*. (IF=8.2) https://doi.org/10.1016/j.eswa.2024.125091
- **G. Liu** et al. (2024) DG-MRM: Dynamic Graph and Multi-view Medication Information for Recommending Medication Combination. Proceedings of *KDD-AIDSH 2024*, AI and Data Science for Healthcare: Bridging Data-centric AI and People-Centric Healthcare workshop. https://openreview.net/forum?id=hdE8YZIAjw

GRANTS

- **Principle Investigator** of the Research on medication recommendation method based on multi-view dynamic graph contrast learning, **National Undergraduate Training Program for Innovation and Entrepreneurship** 05/2023 – 05/2025
- **Principle Investigator** of the Research on interpretable deep learning approach for electronic health records, Shandong Normal University Training Program for Innovation and Entrepreneurship 05/2022 – 05/2023
- **Principle Investigator** of Research on Human-Computer Interaction Methods for NAO Robots, Shandong Normal University undergraduate research fund project 11/2021 – 03/2023

RESEARCH EXPERIENCE

Explainable Deep Learning Methods for Electronic Health Records 05/2023 – 08/2026

Supervisor: Prof. Xiaomei Yu, Shandong Normal University

Work package: Leveraged large language models to analyze medical data in EHR; And accurate drug recommendation methods for electronic health records (EHR) using temporal graph structures.

Outputs includes a medication recommendation software copyright 1.0 registered on 2024.03.29.

Fake News Detection based on Large Language Models 02/2023 – 05/2026

Supervisor: Assoc. Prof. Hongchen Wu, Shandong Normal University

Work package: analyzed semantic interactions between posts and comments in social media datasets to explore how ambiguous event descriptions influence emotional responses to real and fake news; proposed innovative solutions and implemented algorithms.

Outputs include a project won the National Second Prize at the 16th China University Computer Design Competition.

Intelligent Classroom Management System Based on Edge Computing 05/2022 – 05/2023
Supervisor: Dr. Ke Liu, Shandong Normal University
Work package: developed a sound source localization and tracking device using Raspberry Pi 4B and infrared laser technology, designed a contactless temperature measurement, mask detection, and facial recognition system using Raspberry Pi 4B.
Outputs: the project won the 2nd Prize at the 2022 Shandong Province College Student Electronic Design Competition, and achieved First Prize at Shandong Normal University's Electronic Design Competition; Patent granted as the first inventor (2022.10.19, No. 202222749613.8)

Research on Human-Robot Interaction Methods for NAO Robots 07/2021 – 06/2022
Supervisor: Assoc. Prof. Wei Yan, Shandong Normal University
Work package: developed an intelligent obstacle avoidance system for NAO robots using computer vision techniques; designed a music-domain human-robot interaction system using natural language processing and knowledge graph methods.
Outputs: the project won the Third Prize at the 24th China Robotics and Artificial Intelligence Competition (Shandong Province); the Intelligent robot obstacle avoidance V 1.0 system registered national computer software copyright; Patent Granted as the first inventor (2023.10.24， No. ZL 2023 2 0465706.X)

PROFESSIONAL EXPERIENCE

Emergency Management Bureau, Tengzhou, China *Intern*
Responsibilities:
- Dispatched information from the bureau to relevant township and subdistrict officials.
- Assisted in designing and coding a small system for automating duty scheduling.

Shandong Province Artificial Intelligence Education Base, China *Course Developer*
Responsibilities:
- Guided visiting students on research tours, explaining the principles of: 3D printing technology, computer vision-based waste sorting devices, computer vision-based color block sorting devices, computer vision-powered Tic-Tac-Toe robot etc.
- Assisted in the design of a computer vision-based waste management system and color sorting devices using the TensorFlow framework.

SKILLS AND AWARDS

✓ **Technical Skills: Adept at** web design, familiar with the **PyTorch** and **TensorFlow** framework
Programming languages & software: C, C++, Python, Java, MATLAB, and LINGO
Certified in the Industrial and Information Technology Talent Professional Knowledge Assessment – Mathematical Modeling

✓ **Languages:** Mandarin Chinese (Native), English (Fluent, IELTS 6)

✓ **National Prize**
2023- Second Prize, 16th China University Computer Design Competition
2022- Second Prize, **National College Student Mathematical Modeling Competition** *
2022- Third Prize, 12th *MathorCup* Higher Education Mathematical Modeling Challenge
2022- Third Prize, 12th *MathorCup* Higher Education Big Data Competition
2022- Third Prize, 3rd *Huashu Cup* National College Student Mathematical Modeling Competition

✓ **Provincial Prize**
2023- Third Prize, China Collegiate Computing Contest Network Technology Challenge
2022- Second Prize, 13th Blue Bridge Cup National Software and IT Talent Competition
2022- Second Prize, National College Student Electronic Design Competition
2022- Third Prize, 24th China Robotics and Artificial Intelligence Competition